Risk Assessment Plan

rePhlow · iGEM 2026 · Entrepreneurship · TRL 2–3 · as of 30 September 2026

1. Purpose and scope

The iGEM Best Supporting Entrepreneurship criteria ask for a logical product development plan covering milestones, timelines, resources and risks. Risk is therefore one of the four components of the plan rather than a caveat appended to it. This document exists so that the risk component can be examined in the same way as the technical work, with a stated method, stated evidence and stated limits.

The scope covers everything between the current state of the project and the completion of a first industrial pilot. It includes the biological and material system, the reactor, the revalorisation route, the regulatory position, the commercial model, the financial plan and the team itself. It excludes laboratory occupational health and safety, which is governed by the institutional biosafety procedures and the iGEM safety forms, and it excludes the detailed financial model, which is a separate document.

The Business Model Canvas is also a separate document. Each of its nine blocks ends with the register IDs it depends on, and every entry in the register below names the canvas block it bears on, so the two can be read together without repeating each other.

rePhlow is at Technology Readiness Level 2 moving into Level 3. At that maturity most of what the project knows is an assumption rather than a result, and the useful question is not whether risks exist but which of them would invalidate the rest of the work. This document is organised around that distinction.

2. Method

2.1 Framework

The structure follows ISO 31000, which describes risk management as a repeated cycle of identification, analysis, evaluation, treatment and review, carried out on a fixed schedule rather than in response to events [22]. Three departures from a minimal application of that structure are worth stating, because they are what make the register usable rather than decorative.

1. Every risk carries two exposure values, one before treatment and one after, so the register shows what the design decisions are actually buying.
2. Every risk carries an evidence tag recording what its score rests on, so that a well-founded score and a guess are not presented as equivalent.
3. Every risk carries a trigger indicator with a measurable threshold and a pre-agreed action, so that a risk materialising produces a decision rather than a discussion.

2.2 Scoring scales

Likelihood and impact are each scored from 1 to 5 and multiplied to give an exposure value from 1 to 25. Likelihood is assessed over the horizon running to the completion of the industrial pilot. Impact takes the worst of the three dimensions listed, not their average, because a risk that is cheap but fatal to the value proposition is not a moderate risk.

Score	Likelihood over the planning horizon	Impact, worst of cost, schedule and value proposition
1	Rare, below 10 per cent	Absorbed within the current work plan
2	Unlikely, 10 to 30 per cent	Delay under one month, or cost effect under 5 per cent
3	Possible, 30 to 50 per cent	Delay of one to three months, or a closed milestone reopened

Score	Likelihood over the planning horizon	Impact, worst of cost, schedule and value proposition
4	Likely, 50 to 80 per cent	A development phase repeated, or a core assumption invalidated
5	Almost certain, above 80 per cent	The value proposition or the route to market fails

Exposure bands are used throughout the register:

Low 1–4	Medium 5–9	High 10–14	Critical 15–25

A risk matrix is a communication instrument rather than a quantitative model, which is why the bands carry no decision authority on their own and every entry in the top tier also carries a numeric trigger.

2.3 Evidence grading

The water quality laboratory at the Centro de Estudios Hidrográficos made clear during our visit that a preliminary colourimetric method may not be presented as equivalent to an accredited reference method [21]. The team adopted a designed, built, tested and validated standard in response, and this register applies the same discipline to risk scores. Each entry is tagged with the strongest evidence supporting it.

Tag	What it means	Count
Measured	Our own experimental result	1
Literature	Published peer-reviewed work on a comparable system	5
Stakeholder	A documented consultation recorded in Integrated Human Practices	8
Primary law	The text of a directive, regulation or national statute	5
Assumption	Neither data nor a documented source. Reasoned, but unevidenced	11

Eleven entries out of thirty remain assumptions. That is the honest description of a project at this stage, and the register treats reducing that number as an objective in its own right. A risk may not be moved into a lower likelihood band on the strength of an assumption alone.

2.4 Boundary conditions

Three conditions apply to everything in this document and are stated here rather than repeated in each entry.

1. No release of any kind takes place during the competition. iGEM prohibits releasing genetically modified organisms or their products outside the laboratory, explicitly including release inside a containing device [24]. Every statement about field performance in this plan is a design target, and the industrial pilot sits after the competition.
2. All numerical performance values in the business material are development targets rather than results. They are labelled as such wherever they appear.
3. The register describes rePhlow as a prospective company, not as an iGEM team. Risks that only exist after incorporation, such as the environmental liability guarantee, are included because they change decisions being taken now.

3. Risk profile

The register holds thirty risks in seven categories. The table below shows how exposure is distributed before and after treatment.

Category	Risks	Critical before	High before	High or above after
Technical and biological	8	3	5	1

Category	Risks	Critical before	High before	High or above after
Biosafety and containment	4	0	2	0
Regulatory and legal	5	2	1	1
Market and commercial	5	2	2	0
Financial	4	2	2	1
Operational	2	0	1	0
Organisational and reputational	2	0	1	0
Total	30	9	14	3

Nine risks are Critical before treatment and fourteen are High. After treatment none remain Critical and three remain High. Twenty-four of the thirty sit in the Medium band after treatment rather than dropping to Low, because at this maturity most treatments reduce likelihood without removing the underlying uncertainty. A register in which every entry fell to Low would mean the scoring was decorative.

4. The three risks that drive the others

Registers usually present risks as independent entries. In rePhlow they are not. Three sit upstream of most of the rest, and the order in which they are addressed determines whether the plan is realistic. They are set out in full here. Every other entry is held in the register in Section 5, with the same fields in compact form.

TEC-02 — Capsule service life is shorter than the replacement schedule assumes

Evidence: Literature	Inherent 4 × 5 = 20 Critical	Residual 3 × 4 = 12 High	Owner: Materials and hardware

Calcium alginate is a physically crosslinked gel, not a covalent one, and it loses mechanical strength in real wastewater far faster than in buffer. Beads containing immobilised cells lost their mechanical strength after 24 hours of incubation in secondary-treated municipal wastewater while retaining their shape for about 96 hours, and every reinforcement tested by those authors, including polyvinylpyrrolidone, polyvinyl alcohol, carboxymethylcellulose, calcium carbonate and strontium chloride, failed to prevent it [9,10]. Capsule replacement is the largest recurring operating cost in the model, so its frequency is the single most sensitive assumption in the business case.

Treatment. Measure compressive strength, breakage rate and residual activity across repeated cycles in real Bio-Oils effluent rather than in synthetic medium, and report service life as cycles to a defined activity threshold.

Fallback. Treat capsules as a scheduled consumable and price them as a recurring revenue line, so a short service life becomes a business model rather than a defect.

Trigger. More than 20 per cent loss of compressive strength or more than 5 per cent capsule breakage per operating cycle.

REG-02 — A genetically modified process at a customer site requires that site to be an authorised installation

Evidence: Primary law	Inherent 4 × 5 = 20 Critical	Residual 3 × 4 = 12 High	Owner: Entrepreneurship and legal

Encapsulation solves the environmental release question but not the administrative one. Under the contained use regime, activities are classified from type 1 to type 4 by risk, premises must be notified to the competent authority before first use, and types above the lowest require prior authorisation, with the autonomous communities competent for research and development activities [5,6,7]. A customer that installs a reactor containing living modified bacteria becomes the operator of a contained use installation. That is an adoption barrier no amount of encapsulation removes.

Treatment. Confirm with the competent authority whether the unit is classified as contained use at the customer site, keep capsule manufacture inside rePhlow's own notified installation, and quantify the administrative burden a customer would carry.

Fallback. Develop a cell-free variant in which only immobilised enzymes are supplied, which removes the modified organism from the customer site at the cost of the accumulation step.

Trigger. Any prospective customer raising installation authorisation as a blocking condition.

FIN-02 — Cash shortfall between funding cycles

Evidence: Literature	Inherent 4 × 5 = 20 Critical	Residual 3 × 4 = 12 High	Owner: Finance

Running out of capital is the most frequently cited cause of failure among venture backed companies that shut down, appearing in around 70 per cent of analysed cases [20]. rePhlow's income is currently non-recurring grant and prize money with irregular payment calendars, while its costs include a fixed competition commitment. The exposure is the timing gap, not the total.

Treatment. Build a monthly cash projection for the first year and quarterly thereafter, identify the minimum liquidity point explicitly, and phase committed spending so each stage is funded before it begins.

Fallback. Defer capital purchases and reduce the pilot to the smallest configuration that still produces a decision-grade result.

Trigger. Projected cash below three months of committed spend.

4.1 How the risks connect

The dependencies below are the reason the register is sequenced rather than merely sorted.

- TEC-01, TEC-02 and OPS-03 jointly determine the cost of goods, which is FIN-01. FIN-01 in turn drives the cash position, FIN-02. A cost figure published before those three are measured is a forecast presented as a finding.
- TEC-03 and TEC-08 both act on TEC-02, since chemical destabilisation and biological degradation shorten capsule life through different mechanisms and have to be measured separately.
- REG-01 acts on MKT-02. If the regulatory driver is weaker than claimed, the customer compares rePhlow against chemical precipitation on cost alone, which is the comparison rePhlow loses.
- REG-02 acts on the whole commercial model rather than on any single entry, because it changes who the customer has to become in order to buy. In the Business Model Canvas this is why capsule manufacture stays in our own notified facility and why the cell-free module appears as a second configuration.
- REG-04 and MKT-05 act together on Key Resources. Because the project follows the iGEM open model, defensibility has to come from integration, know-how and the supply relationship rather than from exclusive rights, while freedom to operate still has to be checked.

5. Risk register and treatment plan

Exposure is likelihood multiplied by impact. The inherent value assumes no treatment, the residual value assumes the treatment shown is in place and working. Each entry carries a fallback control for the part that cannot be eliminated and a trigger indicator whose threshold converts the risk into an action when it is crossed. The final column names the Business Model Canvas block each entry bears on.

Technical and biological risks

ID	Risk, treatment, fallback and trigger	Evidence and owner	Inherent	Residual	Canvas block
TEC-01	**Encapsulated cells accumulate less phosphorus than free cells.** Sets capsule count, reactor volume and cost per treated cubic metre. **Treatment.**	Assumption — Wet lab, strain	4 × 4 = 16 Critical	3 × 3 = 9 Medium	Cost Structure

ID	Risk, treatment, fallback and trigger	Evidence and owner	Inherent	Residual	Canvas block
	Measure encapsulated and free specific uptake in the same paired assay before any cost figure is published, and carry the ratio as a range rather than a point value through the financial model. **Fallback.** If the ratio falls below 0.6, increase capsule loading and reduce capsule diameter to recover surface area, and re-baseline the reactor sizing rather than the biology. **Trigger.** Encapsulated-to-free specific uptake ratio below 0.6 in the first paired assay.	subgroup			
TEC-02	**Capsule service life is shorter than the replacement schedule assumes.** Replacement is the largest recurring operating cost in the model. **Treatment.** Measure compressive strength, breakage rate and residual activity across repeated cycles in real Bio-Oils effluent rather than in synthetic medium, and report service life as cycles to a defined activity threshold. **Fallback.** Treat capsules as a scheduled consumable and price them as a recurring revenue line, so a short service life becomes a business model rather than a defect. **Trigger.** More than 20 per cent loss of compressive strength or more than 5 per cent capsule breakage per operating cycle.	Literature — Materials and hardware subgroup	4 × 5 = 20 Critical	3 × 4 = 12 High	Revenue Streams, Cost Structure
TEC-03	**The effluent matrix destabilises the calcium alginate network.** Phosphate itself chelates the calcium holding the gel together. **Treatment.** Characterise the real effluent for calcium, phosphate, chelators, conductivity and pH before pilot design, and test capsule integrity across that measured envelope rather than a nominal one. **Fallback.** Maintain a calcium set point in the operating medium and reinforce the shell with a polyelectrolyte coating, accepting the associated mass transfer penalty. **Trigger.** Free calcium in the reactor falling below the value at which capsule swelling begins in the bench test.	Literature — Materials and hardware subgroup	4 × 4 = 16 Critical	3 × 3 = 9 Medium	Key Activities
TEC-04	**Phosphorus speciation in the real stream is not covered by the enzyme panel.** Total phosphorus will not fall if the panel misses the real species. **Treatment.** Fractionate phosphorus in a real Bio-Oils sample into orthophosphate, acid-hydrolysable and organic pools, then match the panel to the measured fractions instead of to a generic profile. **Fallback.** Add a mild chemical or thermal pre-hydrolysis step upstream of the reactor, documented as an added unit operation with its own cost. **Trigger.** Organic and condensed phosphorus above 40 per cent of total phosphorus in the characterised stream.	Stakeholder — Enzyme subgroup	3 × 4 = 12 High	2 × 3 = 6 Medium	Value Propositions
TEC-05	**Enzyme leaching and loss of surface activity across**	Stakeholder —	4 × 3 = 12 High	2 × 3 = 6 Medium	Key Activities

ID	Risk, treatment, fallback and trigger	Evidence and owner	Inherent	Residual	Canvas block
	operating cycles. Attachment at time zero is not retention across cycles. **Treatment.** Quantify protein release and residual activity in the same experiment across cycles, and set a leaching ceiling as an explicit design requirement rather than an outcome. **Fallback.** Move the weakest performers of the panel to covalent attachment, accepting lower initial specific activity in exchange for retention. **Trigger.** Cumulative protein release above the ceiling defined in the immobilisation protocol.	Enzyme subgroup			
TEC-06	**Mass transfer limits performance when the reactor is scaled.** Bench performance is not evidence of packed-basket performance. **Treatment.** Fix basket to reactor volume ratio, packing density and flow pattern against the reference commercial rotating-bed unit inspected during the advisory visit, and validate at two scales before quoting a plant-scale figure. **Fallback.** Increase reactor count rather than reactor size, which preserves the modular architecture and keeps the validated geometry unchanged. **Trigger.** More than 25 per cent drop in volumetric removal rate between the bench and the five litre configuration.	Stakeholder — Hardware subgroup	3 × 4 = 12 High	3 × 3 = 9 Medium	Value Propositions
TEC-07	**The revalorisation cascade is not stable enough to support a product line.** Without a stable cascade rePhlow competes on removal cost alone. **Treatment.** Continue with the wild-type enzyme as the reference case, test a conservative redesigned variant, and keep the product target interchangeable so the cascade is not tied to one molecule. **Fallback.** Sell recovered polyphosphate as an intermediate rather than a finished biomolecule, which lowers the margin but preserves the circular claim. **Trigger.** No variant reaching the reference specific activity after two design iterations.	Measured — Biocatalysis subgroup	3 × 4 = 12 High	3 × 3 = 9 Medium	Revenue Streams
TEC-08	**Effluent microbiota colonise or degrade the capsules.** Industrial effluent is not sterile and alginate degraders are present in it. **Treatment.** Include a non-sterile challenge in the capsule ageing protocol and monitor surface colonisation alongside strength, so degradation and fouling are separated. **Fallback.** Introduce a defined cleaning cycle between batches and treat the associated downtime as a scheduled operating cost. **Trigger.** Visible biofilm or measurable alginate lyase activity in the reactor supernatant.	Literature — Wet lab, strain subgroup	4 × 3 = 12 High	3 × 3 = 9 Medium	Cost Structure

Biosafety and containment risks

ID	Risk, treatment, fallback and trigger	Evidence and owner	Inherent	Residual	Canvas block
BIO-01	**Engineered cells detected in the treated effluent.** One detection event ends the commercial conversation. **Treatment.** Operate a double physical barrier of core-shell encapsulation and basket retention with downstream ultrafiltration, and monitor the effluent by culture and quantitative PCR against a stated limit of detection. **Fallback.** Stop discharge and divert to holding on any positive result, with a pre-agreed inactivation protocol before release. **Trigger.** Any effluent sample above the stated limit of detection by either method.	Assumption — Safety officer	2 × 5 = 10 High	1 × 5 = 5 Medium	Value Propositions
BIO-02	**No accepted quantitative standard exists for physical containment performance.** No agreed metric exists, so we must define and defend our own. **Treatment.** Define a containment metric in cells per litre and gene copies per litre with a stated limit of detection and a stated sampling frequency, and publish the method so it can be challenged. **Fallback.** Report containment as a measured bound with its detection limit rather than as an absolute claim. **Trigger.** Any external reviewer unable to reproduce the containment metric from the published method.	Literature — Safety officer	4 × 3 = 12 High	3 × 3 = 9 Medium	Customer Relationships
BIO-03	**Spent capsules and lysed biomass are genetically modified waste.** Spent capsules are modified waste, not ordinary industrial waste. **Treatment.** Write and validate an inactivation protocol with a defined log reduction, and include waste handling in the operating procedure supplied with the unit. **Fallback.** Return spent capsules to rePhlow for centralised inactivation, keeping the obligation with the manufacturer. **Trigger.** Any batch leaving the site without a completed inactivation record.	Primary law — Safety officer	3 × 3 = 9 Medium	1 × 3 = 3 Low	Customer Relationships
BIO-04	**Horizontal transfer of introduced genetic elements.** Any resistance marker is a regulatory liability in itself. **Treatment.** Move the validated inserts to the chromosome and remove antibiotic resistance markers from the final strain, keeping plasmids for the compatibility testing stage only. **Fallback.** Retain the plasmid-based strain for laboratory work only and declare it as non-commercial. **Trigger.** Any resistance marker still present in the strain proposed for pilot use.	Assumption — Wet lab, strain subgroup	2 × 4 = 8 Medium	1 × 4 = 4 Low	Key Resources

Regulatory and legal risks

ID	Risk, treatment, fallback and trigger	Evidence and owner	Inherent	Residual	Canvas block
REG-01	**The regulation cited as the market driver does not**	Primary law —	4 × 4 = 16 Critical	2 × 3 = 6 Medium	Value Propositions,

ID	Risk, treatment, fallback and trigger	Evidence and owner	Inherent	Residual	Canvas block
	bind the first customer. The Directive governs urban plants, not our first customer. **Treatment.** Re-anchor the commercial argument on the facility's own integrated environmental authorisation and discharge permit, and present the Directive as the direction of European policy rather than as the binding obligation on this customer. **Fallback.** Obtain written specialist legal verification of the obligations applying to the target stream before any figure is used in a pitch. **Trigger.** Any published rePhlow material still stating a 1 milligram per litre limit from 2027.	Entrepreneurship and legal			Customer Segments
REG-02	**A genetically modified process at a customer site requires that site to be an authorised installation.** The customer site becomes a contained use installation. **Treatment.** Confirm with the competent authority whether the unit is classified as contained use at the customer site, keep capsule manufacture inside rePhlow's own notified installation, and quantify the administrative burden a customer would carry. **Fallback.** Develop a cell-free variant in which only immobilised enzymes are supplied, which removes the modified organism from the customer site at the cost of the accumulation step. **Trigger.** Any prospective customer raising installation authorisation as a blocking condition.	Primary law — Entrepreneurship and legal	4 × 5 = 20 Critical	3 × 4 = 12 High	Key Activities, Customer Relationships
REG-03	**Mandatory environmental liability guarantee and insurability of modified organisms.** Strict liability with a mandatory guarantee, cost unknown. **Treatment.** Request written quotations from at least two insurers and treat the premium as a fixed operating line in the cost structure rather than an afterthought. **Fallback.** Restrict early deployments to configurations where rePhlow remains the operator, keeping the liability inside a controlled perimeter. **Trigger.** No insurer willing to quote for the intended configuration.	Primary law — Finance	3 × 3 = 9 Medium	2 × 3 = 6 Medium	Cost Structure, Revenue Streams
REG-04	**Freedom to operate has not been assessed.** Publishing openly does not remove the risk of infringing somebody else's patent. **Treatment.** Commission a freedom to operate search on the reactor configuration and the recovery process, and record the result as a dated document. Because the project follows the iGEM open model, this search is defensive rather than a step towards filing. **Fallback.** Design around any blocking claim, or approach the holder for a licence while the project is still small enough for that conversation to be cheap. **Trigger.** Any blocking claim identified in a preliminary search.	Assumption — Entrepreneurship and legal	3 × 4 = 12 High	2 × 3 = 6 Medium	Key Resources

ID	Risk, treatment, fallback and trigger	Evidence and owner	Inherent	Residual	Canvas block
REG-05	**The regulatory status of the recovered product is undefined.** Revalorisation revenue cannot be booked without a route to market. **Treatment.** Choose the target product with its regulatory route in mind and document the applicable framework alongside the technical specification. **Fallback.** Direct the first products to industrial self-consumption inside the customer's own process, which avoids placing a new substance on the market. **Trigger.** No identified regulatory route for the selected product at the point of pilot design.	Primary law — Entrepreneurship and legal	3 × 3 = 9 Medium	2 × 3 = 6 Medium	Revenue Streams, Channels

Market and commercial risks

ID	Risk, treatment, fallback and trigger	Evidence and owner	Inherent	Residual	Canvas block
MKT-01	**Dependence on a single industrial partner.** Design reference and commercial reference are the same company. **Treatment.** Advance the discussions already open with a second industrial operator and a public water body, and characterise at least one additional effluent type so the design is not tuned to a single stream. **Fallback.** Use a defined semi-synthetic stream as the reference case so development can continue without plant access. **Trigger.** More than one quarter without contact or sample supply from the reference partner.	Stakeholder — Entrepreneurship	4 × 4 = 16 Critical	3 × 3 = 9 Medium	Key Partners, Customer Segments
MKT-02	**Chemical precipitation remains cheaper on a pure compliance basis.** On compliance alone, precipitation wins on cost. **Treatment.** Build the value argument on avoided sludge management and recovered product value, and quantify both from the customer's own figures rather than from generic market data. **Fallback.** Position the unit as a complement to existing precipitation rather than a replacement, which lowers the switching decision the customer has to make. **Trigger.** Customer valuation of recovered product below the incremental operating cost of the unit.	Stakeholder — Entrepreneurship	4 × 4 = 16 Critical	3 × 3 = 9 Medium	Value Propositions
MKT-03	**The price point and the assumed gross margin are unvalidated.** A margin assumed is a forecast, not a finding. **Treatment.** Rebuild the cost of goods twice, at laboratory prices and at bulk industrial prices, and test the price against the customer's avoided cost rather than against competitor list prices. **Fallback.** Shift to a pay per use or treated volume model, which moves the capital decision away from the customer and lengthens the revenue tail. **Trigger.** Bottom-up cost of goods exceeding 60 per cent of the assumed sale price.	Assumption — Finance	4 × 3 = 12 High	3 × 3 = 9 Medium	Revenue Streams

ID	Risk, treatment, fallback and trigger	Evidence and owner	Inherent	Residual	Canvas block
MKT-04	**A customer refuses modified organisms regardless of the containment evidence.** Some customers decline on policy rather than on evidence. **Treatment.** Prepare the containment dossier as a standard commercial document rather than as a response to objections, and qualify prospects on their organism policy early in the sales conversation. **Fallback.** Offer the cell-free enzymatic configuration to customers whose policy excludes living modified organisms. **Trigger.** Two or more qualified prospects citing organism policy as the reason for declining.	Stakeholder — Entrepreneurship	3 × 4 = 12 High	2 × 4 = 8 Medium	Customer Relationships
MKT-05	**Incumbents or customers internalise the recovery step.** Incumbents already own recovery products and the client networks. **Treatment.** Because the parts are open, build defensibility on the integrated system, on operating know-how and on switching cost through the monitoring platform and the capsule supply relationship. **Fallback.** Licence the technology to an operator rather than compete with one, converting a competitive threat into a channel. **Trigger.** An incumbent announcing an integrated capture and recovery module for variable industrial effluent.	Assumption — Entrepreneurship	3 × 3 = 9 Medium	2 × 3 = 6 Medium	Key Resources, Channels

Financial risks

ID	Risk, treatment, fallback and trigger	Evidence and owner	Inherent	Residual	Canvas block
FIN-01	**Cost of goods materially above the assumed figure.** Inherits every technical uncertainty upstream of it. **Treatment.** Model cost of goods as a range driven explicitly by the three technical parameters, and publish the sensitivity rather than a single figure. **Fallback.** Reprice or restructure toward a service model in which the capsule cost is recovered over the contract term. **Trigger.** Any of the three driving parameters landing outside the modelled range.	Assumption — Finance	4 × 4 = 16 Critical	3 × 3 = 9 Medium	Cost Structure
FIN-02	**Cash shortfall between funding cycles.** The exposure is the timing gap, not the total. **Treatment.** Build a monthly cash projection for the first year and quarterly thereafter, identify the minimum liquidity point explicitly, and phase committed spending so each stage is funded before it begins. **Fallback.** Defer capital purchases and reduce the pilot to the smallest configuration that still produces a decision-grade result. **Trigger.** Projected cash below three months of committed spend.	Literature — Finance	4 × 5 = 20 Critical	3 × 4 = 12 High	Cost Structure

ID	Risk, treatment, fallback and trigger	Evidence and owner	Inherent	Residual	Canvas block
FIN-03	**The pilot investment figure is top-down rather than built up.** A round number that cannot be traced is worth less than a range that can. **Treatment.** Rebuild the figure from equipment quotations, installation, consumables, labour, regulatory analysis and insurance, and state what changed relative to the earlier number. **Fallback.** Present the pilot budget as a range with named scope options rather than as a single committed figure. **Trigger.** More than 25 per cent divergence between the bottom-up total and the published figure.	Assumption — Finance	4 × 3 = 12 High	2 × 3 = 6 Medium	Cost Structure
FIN-04	**Funding is competitive, non-recurring and calendar bound.** One unsuccessful call moves a milestone by a quarter. **Treatment.** Maintain a rolling calendar of at least three funding routes per period and submit ahead of the milestone that depends on them. **Fallback.** Reduce scope to the experiments that unblock the next decision, deferring anything that only improves confidence. **Trigger.** Two consecutive unsuccessful applications in the same period.	Assumption — Finance	4 × 3 = 12 High	3 × 3 = 9 Medium	Cost Structure

Operational risks

ID	Risk, treatment, fallback and trigger	Evidence and owner	Inherent	Residual	Canvas block
OPS-02	**Dependence on advisors and borrowed laboratory capacity.** Equipment access becomes critical path once a pilot is scheduled. **Treatment.** Schedule equipment-dependent work against confirmed availability and identify a second route for each critical measurement. **Fallback.** Outsource the specific measurement to a service laboratory and carry the cost. **Trigger.** Any critical measurement with a single available instrument and no alternative.	Assumption — Coordination	3 × 3 = 9 Medium	2 × 3 = 6 Medium	Key Resources, Key Partners
OPS-03	**Supply and cost of recombinant enzymes at kilogram scale.** Seven enzymes at kilogram scale, currently unpriced. **Treatment.** Request indicative quotations at pilot quantity and evaluate in-house production for the two enzymes contributing most activity. **Fallback.** Reduce the panel to the smallest set that covers the measured speciation of the target stream. **Trigger.** Quoted enzyme cost above 20 per cent of the unit cost of goods.	Assumption — Enzyme subgroup	3 × 4 = 12 High	3 × 3 = 9 Medium	Cost Structure, Key Partners

Organisational and reputational risks

ID	Risk, treatment, fallback and trigger	Evidence and owner	Inherent	Residual	Canvas block
ORG-01	**Overclaiming in public material.** One unsupported claim damages the supported ones beside it. **Treatment.** Label every quantitative statement as designed, built, tested or validated, and mark development targets as targets wherever they appear. **Fallback.** Correct publicly and state the correction, which costs less than defending an untenable figure. **Trigger.** Any published figure that cannot be traced to a method or a source.	Stakeholder — All subgroups	3 × 4 = 12 High	1 × 4 = 4 Low	Value Propositions
ORG-02	**Public opposition to modified organisms in water treatment.** Public acceptance cannot be assembled after opposition forms. **Treatment.** Treat plain language communication of containment and of what the system does as a continuing part of the implementation route rather than as an outreach activity. **Fallback.** Engage early with local stakeholders at any prospective pilot site before the installation is announced. **Trigger.** Organised local objection at a prospective site.	Stakeholder — Human practices	3 × 3 = 9 Medium	2 × 3 = 6 Medium	Channels, Customer Segments

6. Regulatory position in detail

6.1 What Directive (EU) 2024/3019 actually requires

The recast urban wastewater treatment Directive sets total phosphorus at 0.5 milligrams per litre for treatment plants above 150000 population equivalents from 2039, and 0.7 milligrams per litre for plants above 10000 population equivalents from 2045, and it lowers the general coverage threshold to agglomerations above 1000 population equivalents. Member states must transpose it into national law by mid 2027, which is the date the earlier figure appears to have come from [1,2].

The Directive also concerns urban wastewater. Bio-Oils Huelva is an industrial operator whose discharge conditions are set in its own integrated environmental authorisation, granted under the integrated pollution prevention and control regime, rather than by the urban Directive [3,4]. Presenting the Directive as the compliance trigger for that specific customer is the most exposed claim in the entrepreneurship package, and it is corrected in the current version of the wiki.

The corrected version remains a strong commercial argument. European policy is moving toward lower nutrient discharge and higher recovery, phosphate rock and white phosphorus remain on the European list of critical raw materials although they are not designated strategic [23], and some member states already impose recovery obligations directly. Germany requires phosphorus recovery from sewage sludge from January 2029 for treatment plants above 100000 population equivalents and from January 2032 above 50000, where the sludge dry matter contains 2 per cent phosphorus or more [25]. What must change is the precision of the claim, not the direction of the argument.

6.2 Contained use at a customer site

Encapsulation answers the question a large industrial adopter asked, which was whether environmental release could be prevented in normal operation and under foreseeable failure. It does not answer the administrative question, and the two are frequently conflated.

Under the European contained use regime and its Spanish transposition, activities involving genetically modified micro-organisms are classified into four types by risk. Premises must be notified to the competent authority before first use,

activities of null or insignificant risk require notification only, and everything above that requires prior authorisation, with the autonomous communities competent for research and development activity and the National Biosafety Commission providing the mandatory risk evaluation [5,6,7]. A customer that installs a reactor containing living modified bacteria therefore becomes the operator of a contained use installation, with the notification, containment, record-keeping and waste obligations that follow.

This is an adoption barrier that no improvement in encapsulation removes, and it is the reason the register scores REG-02 at the top of the regulatory category. Two consequences follow for the design. Capsule manufacture stays inside rePhlow's own notified installation rather than at the customer site. And the enzyme module is kept separable from the bacterial one, so that a cell-free configuration supplying only immobilised phosphatases remains available for customers who cannot or will not host modified organisms. That configuration loses the accumulation step and most of the recovery value, which is why it is a fallback rather than the plan.

6.3 Liability, insurance and the recovered product

Spanish environmental liability law places activities involving genetically modified organisms under a strict liability regime with a mandatory financial guarantee for the operators it covers [8]. Neither the premium nor the availability of that cover for a small company working with modified bacteria is currently known, and it was already flagged as a gap in the team's financial notes without being quantified. Written quotations from at least two insurers are required before the pilot budget can be considered complete.

The recovered phosphorus product is a regulated article the moment it is sold. Sold as a fertilising product it falls under the European fertilising products regulation, sold as a chemical it carries registration obligations, and a product derived from a modified organism raises separate traceability questions [17,18]. Directing the first output to industrial self-consumption inside the customer's own process avoids placing a new substance on the market and is the lowest friction route for a first pilot.

6.4 Containment evidence and its limits

Genetic biocontainment is assessed through escape frequency, and the strongest engineered systems report undetectable escape at a detection limit near two parts in ten to the twelve [15]. Physical containment has no equivalent agreed metric, and a recent analysis concludes that inconsistent detection limits and the absence of standardised test conditions make containment claims difficult to compare between systems at all [14]. rePhlow cannot borrow a threshold from that literature.

The position taken here is to define a containment metric in cells per litre and gene copies per litre, to state its limit of detection and sampling frequency, and to publish the method so that it can be challenged. The chassis itself is a documented safety strain, certified as a host-vector system 1 organism, a status that is often misreported and should be cited accurately [16]. The final strain will carry no antibiotic resistance markers and the validated inserts will be chromosomal, which removes the most credible horizontal transfer pathway rather than arguing that it is improbable.

6.5 Intellectual property under the iGEM open model

rePhlow follows iGEM's open model. Parts are shared through the Registry of Standard Biological Parts, and the wiki is published under the Creative Commons Attribution 4.0 licence, which permits sharing and adaptation for any purpose including commercial use, requires only attribution, and is irrevocable [27]. iGEM recognises that teams may wish to protect their work and directs them to their institution's technology transfer office, which for rePhlow is the UCM OTRI [26].

The consequence for this plan is a matter of sequence rather than of permission. In Europe there is no grace period: an invention disclosed on the wiki or at the Jamboree is generally no longer novel and therefore no longer patentable, whereas the United States allows a filing within one year of public disclosure. Any decision to protect something has to be taken with the OTRI before the wiki freeze, not after it.

Two entries in the register follow from this position. REG-04 keeps freedom to operate as a defensive check, because publishing openly does not protect rePhlow from infringing somebody else's patent. MKT-05 places defensibility in the integrated system, the operating know-how and the capsule supply and monitoring relationship rather than in exclusive rights over parts.

7. Monitoring, ownership and review

The register is reviewed at each Design–Build–Test–Learn iteration and at every phase boundary of the execution plan. Each review either changes a score, changes an evidence tag, or records explicitly that neither changed. New evidence moves a risk up as readily as down, and a review that produces no change is still recorded, because the absence of a record is indistinguishable from the absence of a review.

Owner	Entries held
Wet lab, bacteria subgroup	TEC-01, TEC-08, BIO-04
Hardware subgroup	TEC-02, TEC-03, TEC-06
Enzyme subgroup	TEC-04, TEC-05, OPS-03
Biocatalysis subgroup	TEC-07
Safety	BIO-01, BIO-02, BIO-03
Entrepreneurship and legal	REG-01, REG-02, REG-04, REG-05, MKT-01, MKT-02, MKT-04, MKT-05
Finance	REG-03, MKT-03, FIN-01, FIN-02, FIN-03, FIN-04
Coordination	OPS-02
Human practices	ORG-02
All subgroups	ORG-01

8. Residual risk and what this plan does not claim

Three statements are made explicitly, because a risk assessment that appears to resolve everything is not an assessment.

1. The residual scores are not zeros. Twenty-four entries remain in the Medium band after treatment. At Technology Readiness Level 3 that is the correct result rather than a shortcoming.
2. Three entries remain High after treatment. Capsule service life, contained use authorisation at the customer site and the cash timing gap cannot be reduced further by design decisions alone. Each of them requires either an experimental result or an external answer that the project does not yet have.
3. Eleven entries rest on assumptions. Converting those assumptions into measurements is the single most useful thing the project can do for the credibility of its business case, and it is why the first three experiments in the execution plan are the paired uptake assay, the capsule ageing series in real effluent, and the characterisation of the Bio-Oils stream.

The strongest claim this plan makes is not that rePhlow is low risk. It is that rePhlow knows, in specific and measurable terms, what would have to be true for the project to work, and what it will do in each case where it turns out not to be.

9. References

[1] European Parliament and Council of the European Union (2024). Directive (EU) 2024/3019 concerning urban wastewater treatment (recast). Official Journal of the European Union, 12 December 2024. https://eur-lex.europa.eu/eli/dir/2024/3019/oj/eng

[2] Federal Ministry of Agriculture and Forestry, Climate and Environmental Protection, Regions and Water Management, Austria (2025). Municipal Wastewater Directive (Directive 2024/3019).

[3] Junta de Andalucía (2017). Autorización Ambiental Integrada, Bio-Oils Huelva, planta de biodiésel La Rábida, Palos de la Frontera, Huelva. Expediente AAI/HU/075.

[4] European Parliament and Council of the European Union (2010). Directive 2010/75/EU on industrial emissions (integrated pollution prevention and control). OJ L 334/17.

[5] European Parliament and Council of the European Union (2009). Directive 2009/41/EC on the contained use of genetically modified micro-organisms (recast). OJ L 125/75.

[6] Jefatura del Estado, España (2003). Ley 9/2003, de 25 de abril, por la que se establece el régimen jurídico de la utilización confinada, liberación voluntaria y comercialización de organismos modificados genéticamente, and Real Decreto 178/2004, de 30 de enero. BOE.

[7] Instituto Nacional de Seguridad y Salud en el Trabajo (2024). NTP 1201. Riesgo biológico — utilización confinada de organismos modificados genéticamente. INSST, Madrid.

[8] Jefatura del Estado, España (2007). Ley 26/2007, de 23 de octubre, de Responsabilidad Medioambiental. BOE-A-2007-18475.

[9] Cruz, I., Bashan, Y., Hernàndez-Carmona, G., & de-Bashan, L. E. (2013). Biological deterioration of alginate beads containing immobilized microalgae and bacteria during tertiary wastewater treatment. Applied Microbiology and Biotechnology, 97(22), 9847–9858. https://doi.org/10.1007/s00253-013-4703-6

[10] Smidsrød, O., & Skjåk-Bræk, G. (1990). Alginate as immobilization matrix for cells. Trends in Biotechnology, 8(3), 71–78. https://doi.org/10.1016/0167-7799(90)90139-O

[11] Zheng, Y., et al. (2023). Recovery of phosphorus from wastewater. Critical Reviews in Environmental Science and Technology, 53(11), 1148–1172. https://doi.org/10.1080/10643389.2022.2128194

[12] Sheldon, R. A., & van Pelt, S. (2013). Enzyme immobilisation in biocatalysis — why, what and how. Chemical Society Reviews, 42(15), 6223–6235. https://doi.org/10.1039/c3cs60075k

[13] Advisory consultations with Juan Manuel Bolívar and Victoria Eugenia Santos, Department of Chemical Engineering, Faculty of Chemistry, UCM, on rotating-basket reactor geometry, packing density and shear. See Integrated Human Practices.

[14] George, D. R., Danciu, M., Davenport, P. W., Lakin, M. R., Chappell, J., & Frow, E. K. (2024). A bumpy road ahead for genetic biocontainment. Nature Communications, 15, 650. https://doi.org/10.1038/s41467-023-44531-1

[15] Mandell, D. J., Lajoie, M. J., Mee, M. T., et al. (2015). Biocontainment of genetically modified organisms by synthetic protein design. Nature, 518, 55–60. https://doi.org/10.1038/nature14121

[16] Kampers, L. F. C., Volkers, R. J. M., & Martins dos Santos, V. A. P. (2019). Pseudomonas putida KT2440 is HV1 certified, not GRAS. Microbial Biotechnology, 12(5), 845–848. https://doi.org/10.1111/1751-7915.13443

[17] European Parliament and Council of the European Union (2019). Regulation (EU) 2019/1009 laying down rules on the making available on the market of EU fertilising products. OJ L 170/1.

[18] European Parliament and Council of the European Union (2006). Regulation (EC) No 1907/2006 concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals (REACH). OJ L 396/1.

[19] Consultations with Canal de Isabel II and IMDEA on the persistence of conventional treatment infrastructure and on integration into facilities not designed for recovery. See Integrated Human Practices.

[20] CB Insights (2026). Why startups fail, top reasons. Analysis of 431 venture-backed shutdowns since 2023. https://www.cbinsights.com/research/report/startup-failure-reasons-top/

[21] Consultation with Carlos López Monllor and Almudena Domínguez Cabrerizo, Centro de Estudios Hidrográficos, CEDEX. See Integrated Human Practices.

[22] International Organization for Standardization (2018). ISO 31000:2018, Risk management. Guidelines. https://www.iso.org/standard/65694.html

[23] European Parliament and Council of the European Union (2024). Regulation (EU) 2024/1252 establishing a framework for ensuring a secure and sustainable supply of critical raw materials.

[24] iGEM Foundation. Safety policy, do not release. https://responsibility.igem.org/safety-policies/release-beyond-containment

[25] Bundesrepublik Deutschland. Klärschlammverordnung (AbfKlärV). Phosphorus recovery obligation from 1 January 2029 for treatment plants above 100000 population equivalents and from 1 January 2032 above 50000, where sludge dry matter contains 2 per cent phosphorus or more.

[26] iGEM Foundation. Frequently Asked Questions: what if we want to patent our work? https://competition.igem.org/participation/faq

[27] iGEM Foundation. Team Wiki requirements and Creative Commons Attribution 4.0 licensing of wiki content.